

Rajat Savdekar

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EDUCATION

University of Southern California, Los Angeles, CA Master of Science in Computer Science Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, VR / XR Development	Aug 2024 – May 2026 GPA: 3.63
University of Mumbai, Mumbai, India Bachelor of Engineering in Computer Engineering Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing	Aug 2019 – May 2023 CGPA: 9.4/10

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Java
AI & Agents: OpenAI API, LangChain, LangGraph, RAG, vector search, FAISS, Whisper, PyTorch, model quantization
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets, Celery
Data & Storage: PostgreSQL, MongoDB, Redis, LevelDB, Yjs CRDTs
Cloud & Infrastructure: AWS, Docker, Terraform, GitHub Actions, CI/CD
Frontend: React, Next.js, Tailwind CSS

EXPERIENCE

Software Engineer Easley-Dunn Productions, Inc. – Mediverse (volunteer)	Aug 2026 – Present Torrance, CA
- Cut replay narration turnaround from 22 minutes to 4 by keeping a 4-bit quantized 7B model resident instead of cold-loading per request, with batched speech synthesis and content-hash caching. - Built a human-in-the-loop gate that validates each narration sentence against the event log before release. - Raised replay upload success from 82% to 99.6% by moving 100 MB blobs out of SQLite into streamed object storage.	
Teaching Assistant USC Summer Programs	May 2025 – Jul 2025 Los Angeles, CA
- Taught Python, Git, and REST API design to roughly 150 students across three USC summer cohorts. - Ran daily code review and debugging sessions, taking 20+ teams from spec to a working hardware-integrated demo.	
Software Engineer – Backend & Distributed Systems Jalgaon Fruit Sales Cooperative Ltd (JFSS)	May 2023 – May 2024 Jalgaon, India
- Doubled peak harvest write throughput by architecting an isolated write path that eliminated the 10–15 minute MyISAM table-lock freezes stalling the platform every season, sustaining growth from \$100K to \$200K in transaction volume. - Decomposed a legacy PHP monolith into modular Python Flask microservices on PostgreSQL, running both paths in parallel with cron-based state sync until cutover, cutting query latency 40% across 1,000+ daily truck transactions. - Delivered a web procurement platform with REST APIs and live analytics dashboards, migrating 150+ daily clerks across 5 warehouses off manual logbooks.	
Software Engineer Intern Softaid Computers	Jan 2023 – Apr 2023 Jalgaon, India
- Evaluated domain fine-tuning against retrieval-augmented generation for support queries, selecting RAG after weighing answer traceability and index refresh cost against the retraining cycle. - Built the ingestion pipeline that chunked, embedded, and indexed the support ticket archive into a FAISS index.	

PROJECTS

MockPad – Real-Time Collaborative Code Editor Node.js, React, WebSockets, Yjs CRDTs, OpenAI Whisper, MongoDB, LevelDB, Python	[Live] [GitHub] May 2026
- Built a real-time collaborative editor on Yjs CRDTs over a custom WebSocket layer, reconciling concurrent edits without a central lock and restoring a session cleanly when a collaborator reconnects. - Evicts idle rooms to LevelDB on an LRU policy, so the whole thing runs on one \$6/month droplet with no lost state.	
Caliberate – LLM Code-Evaluation Sandbox Python, FastAPI, PostgreSQL, Docker, GitHub Actions	[GitHub] Aug 2026
- Runs model patches in isolated Docker containers off a PostgreSQL SKIP LOCKED queue, with leases for crash recovery. - Wired into CI to run the regression suite against each patch and reject the ones that break it.	

RESEARCH & HACKATHONS

Publications. Four undergrad papers (IJNRD, IJIRCCE, 2022–23); *Texture Feature Analysis Using GLCM Matrix* shipped a MATLAB feature-extraction tool since cited by 10+ papers in IEEE and Springer volumes. [GitHub]
Aura – visionOS Accessibility Assistant. 2nd place of 40+ teams, USC 24-hour hackathon, Aug 2025. [Demo]
Stitch – Autonomous CI Repair Agent. OpenAI Build Week, Developer Tools track, Aug 2026. [GitHub]